

Machine Learning Notes for RAG Assignment

Introduction to Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables systems to learn patterns from data and improve performance without being explicitly programmed.

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Supervised Learning

Supervised Learning uses labeled datasets to train models. Common algorithms include Linear Regression, Logistic Regression, Decision Trees, and Random Forest.

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Unsupervised Learning

Unsupervised Learning finds hidden patterns in unlabeled data. Clustering and dimensionality reduction are common techniques.

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Classification

Classification predicts categorical outputs such as spam or not spam, disease or no disease.

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Regression

Regression predicts continuous values such as house prices, sales forecasts, and temperature.

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Decision Trees

Decision Trees split data based on feature values and are easy to interpret for prediction tasks.

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Random Forest

Random Forest combines multiple decision trees to improve accuracy and reduce overfitting.

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Model Evaluation

Accuracy, Precision, Recall, F1-Score, and ROC-AUC are common metrics used to evaluate machine learning models.

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